N+ Network Security

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Assignment-11

Prepare By
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1. **Install and set up a DNS server on your local machine or a VM using BIND (Linux) or Simple DNS Plus (Windows), then create a new DNS zone for the domain musicmania.in.**

Aim:

To install and configure a DNS server using Simple DNS Plus on Windows and create a new DNS zone named musicmania.in.

Procedure:

1. Downloaded and installed Simple DNS Plus on Windows.
2. Opened the Simple DNS Plus application.
3. Created a new Primary DNS Zone named musicmania.in.
4. Added an A Record to map the domain name to an IP address (192.168.1.10).
5. Added a CNAME Record for www.musicmania.in.
6. Tested the DNS server using the nslookup command.

Result:

Successfully installed and configured the DNS server using Simple DNS Plus and created a new DNS zone named musicmania.in.

2. **Add an A record in your DNS server configuration to map www.musicmania.in to the IP address 192.168.1.50, then use the nslookup or dig command to verify the mapping works.**

Aim:

To add an A Record in the DNS server and map www.musicmania.in to 192.168.1.50.

Procedure:

1. Opened Simple DNS Plus.
2. Selected the musicmania.in DNS Zone.
3. Created a new A Record.
4. Entered the hostname www.
5. Entered the IP address 192.168.1.50.
6. Saved the DNS Record.
7. Opened CMD and executed:

nslookup www.musicmania.in 127.0.0.1

Expected Output:

Name: www.musicmania.in

Address: 192.168.1.50

Result:

Successfully created an A Record for www.musicmania.in pointing to 192.168.1.50 and verified it using the nslookup command.

3. **Simulate a typo by adding a CNAME record for play.musicmania.in that points to www.musicmania.in, then test if accessing play.musicmania.in resolves to the correct IP address.**

Aim:

To create a CNAME Record for play.musicmania.in pointing to www.musicmania.in and test DNS resolution.

Procedure:

1. Opened Simple DNS Plus.
2. Selected the musicmania.in DNS Zone.
3. Created a new CNAME Record.
4. Entered the hostname play.
5. Set the target hostname to www.musicmania.in.
6. Saved the DNS Record.
7. Opened CMD and executed:

nslookup play.musicmania.in 127.0.0.1

Expected Output:

Name: www.musicmania.in

Address: 192.168.1.50

Aliases: play.musicmania.in

Result:

Successfully created a CNAME Record for play.musicmania.in pointing to www.musicmania.in. Verified that it resolves to the same IP address 192.168.1.50

4. **Your DNS server is not resolving the domain musicmania.in as expected. Document the step-by-step troubleshooting process you would follow, including checking configuration files, restarting the DNS service, and testing with nslookup or dig.Hint: Include at least three different troubleshooting commands or checks in your documentation.**

Aim:

To troubleshoot DNS resolution problems for musicmania.in.

Troubleshooting Steps:

1. **Check DNS Configuration:** Open Simple DNS Plus and verify that the musicmania.in zone and DNS records are configured correctly.
2. **Check IP Configuration:** Open CMD and execute:

ipconfig /all

Check the DNS Server IP address.

3. **Check DNS Service:** Open Windows Services (services.msc) and check whether the Simple DNS Plus service is running. Restart the service if required.
4. **Flush DNS Cache:** Execute:

ipconfig /flushdns

This clears the local DNS cache.

5. **Test DNS Resolution:** Execute:

nslookup musicmania.in 127.0.0.1

Check whether the correct IP address is returned.

6. **Test Network Connectivity:** Execute:

ping 192.168.1.100

Check whether the DNS server is reachable. (Use the actual DNS server IP.)

7. **Check DNS Port:** Verify that Windows Firewall allows DNS traffic on UDP and TCP port 53.

Result:

Successfully identified and troubleshot DNS configuration issues by checking DNS records, restarting the DNS service, clearing the DNS cache, and testing DNS resolution using nslookup.